



Transformer Reliability Survey

Tutorial of CIGRÉ WG A2.37

Convener: Stefan Tenbohlen, Germany



AIM

- CIGRE (**International Council on Large Electric Systems**) is one of the leading worldwide Organizations on Electric Power Systems, covering their technical, economic, environmental, organisational and regulatory aspects.
 - > Facilitate the exchange of information between engineering personnel and specialists in all countries and develop knowledge in power systems.
 - > Add value to the knowledge and information exchanged by synthesizing state-of-the-art world practices.
 - > Make managers, decision-makers and regulators aware of the synthesis of CIGRE's work, in the area of electric power.

More specifically, issues related to planning and operation of power systems, as well as design, construction, maintenance and disposal of HV equipment and plants are at the core of CIGRE's mission. Problems related to protection of power systems, telecontrol, telecommunication equipment and information systems are also part of CIGRE's area of concern.

05.04.2011

Cigre Study Committees

- A1 Rotating Electrical Machines
- A2 Transformers
- A3 High Voltage Equipment
- B1 Insulated Cables
- B2 Overhead Lines
- B3 Substations
- B4 HVDC and Power Electronics
- B5 Protection and Automation
- C1 System Development and Economics
- C2 System Operation and Control
- C3 System Environmental Performance
- C4 System Technical Performance
- C5 Electricity Markets and Regulation
- C6 Distribution Systems and Dispersed Generation
- D1 Materials and Emerging Test Techniques
- D2 Information Systems and Telecommunication

05.04.2011



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Outline

1. MOTIVATION AND TERMS OF REFERENCE
2. THEORETICAL BACKGROUND AND DEFINITIONS
3. DESCRIPTION OF EXISTING TRANSFORMER RELIABILITY SURVEYS
4. DEVELOPED METHODOLOGY FOR STANDARDIZED FAILURE DATA COLLECTION
5. RESULTS OF PERFORMED RELIABILITY SURVEY
6. CONCLUSION AND RECOMMENDATION

Motivation

Transformer Reliability Survey



- Accurate information about service experience of high voltage equipment is of significant value for both electric utilities and for manufacturers,
- It helps the manufacturers to improve their products,
- It provides important inputs for the utilities when buying equipment, when organizing maintenance and when benchmarking their performance,

$$\text{Total Cost of Ownership} = \text{First Price} + \text{Cost of Losses} + \text{Cost of not Running}$$

- Statistical analysis of the past failure data can display useful features with respect to the future failure behavior,
- Equipment reliability data are also required when assessing the overall reliability of an electric power system,
- Furthermore, international standards applicable to high voltage equipment are being improved on the basis of service experience and reliability data.

Life Assessment Methodologies

Top-down Analysis

- Investigation of failure behavior in the whole population of assets by means of analytical tools (e.g. statistical distributions),
- Information about number and ages of failed and installed units is necessary,
- Emphasis on economic and strategic life-time assessment,
- Outputs are e.g. failure frequency, age of assets which are most likely to fail. Results can be used to parameterize Time Based Maintenance.

Bottom-up Analysis

- Degradation and condition assessment of individual assets based on
 - loading history, aging characteristics
 - maintenance and diagnostic reports (e.g. DGA, PD, FRA, Moisture)
 - post-mortem investigation.

*L. Chmura, „Life-Cycle assessment of high-voltage assets using statistical tools“, PhD thesis, Delft, 2014

Terms of Reference, WG A2.37

Transformer Reliability Survey



The current CIGRE reliability statistic is/was based on failure data from 1968 to 1978! *

- Several approaches failed because of restriction to data access

Terms of Reference:

- Review all existing national surveys and study different practices (data collection, compilation, etc.)
- Conducting a new international survey on transformer failures, and proposing a uniform way of collecting, compiling and presenting data.
- Compiling and analysing the collected data, and interpreting the results (calculation of failure rates, classification into failure location, failure causes and failure modes)
- Recommendations

*A. Bossi, e. al, „An international survey on failures in large power transformers in service“ *Cigré Electra* No.88, 1983.

THEORETICAL BACKGROUND AND DEFINITIONS

Definitions

Failure

- Any situation which requires the equipment to be removed from service to be repaired (Cigre WG A2.18, 2003).
- The systems operator's focus would be on the impact on the system, ranking failure in terms of system reliability, whereas the plant specialist would rank it in terms of what remedial action would be required to restore equipment functionality

Reliability

- Ability of an item to perform a required function under given conditions for a given time interval (IEC, 1986)
- Probability that the equipment will remain in service without a failure occurring (Cigre WG A2.18, 2003).

Failure Rate

- Number of failures divided by the number of transformers in service over a period of time.

Failure rate of a single population:

$$\lambda = 100 \% \cdot \frac{n_1 + n_2 + \dots + n_i}{(N_1 + N_2 + \dots + N_i) \cdot T}$$

n_i is the number of failures in the i-th year

N_i is the number of transformers operating in the i-th year

T is the reference period (one year)

Failure rate of combined population:

$$\lambda = 100 \% \cdot \frac{n_1 + n_2 + \dots + n_i}{N_1 \cdot T_1 + N_2 \cdot T_2 + \dots + N_i \cdot T_i}$$

n_i is the number of failures by i-th population

N_i is the number of transformers of i-th population

T_i is the reference period of i-th population

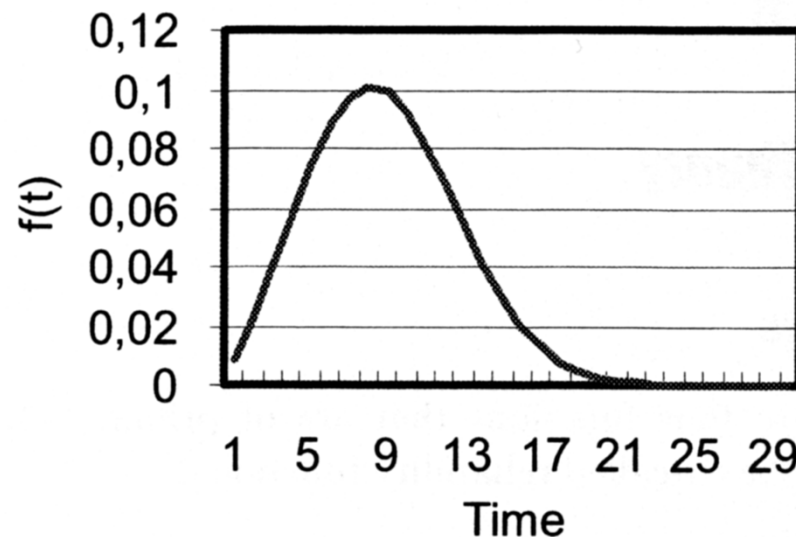
Definitions

Probability Density Function (PDF)

- The probability density function (PDF), $f(t)$, gives the probability of failure occurring at any specific time. Its units are failure/item-time. Thus, at any point it gives the probability of a failure occurring during the following time interval.

$$f(T) = \frac{1}{N} \cdot \frac{\Delta n}{\Delta t}$$

Δn = Number of failures in time interval $[t, t+\Delta t]$
 Δt = Length of time interval
 N = Original Population

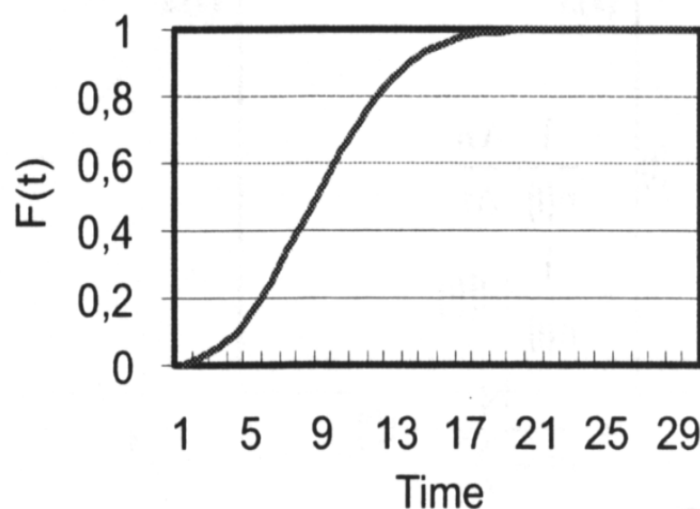


Definitions

Failure Distribution Function (CDF)

- The failure distribution function is a *cumulative distribution function* and gives the cumulative probability of failure. It thus represents the probability that a failure has occurred on or before a certain time.

$$F(t) = \int_0^t f(t)dt = \frac{\sum n_i(t)}{N} \quad \begin{array}{l} \sum n_i = \text{Number of failures up to time } t \\ N = \text{Original Population} \end{array}$$

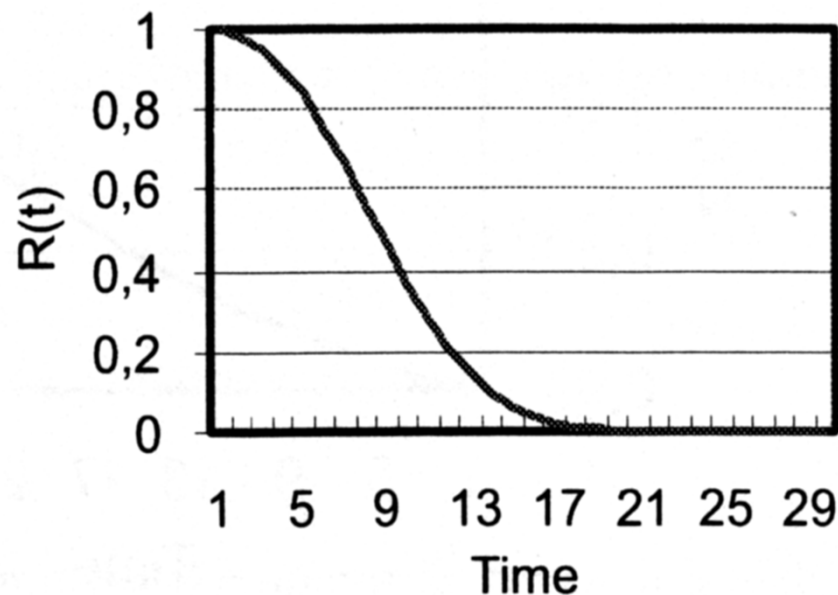


Definitions

Survival (or Reliability) Function

- The survival (reliability) function is the complement of the failure distribution function. It gives the probability of survival up to any specific time t .

$$R(t) = 1 - F(t)$$



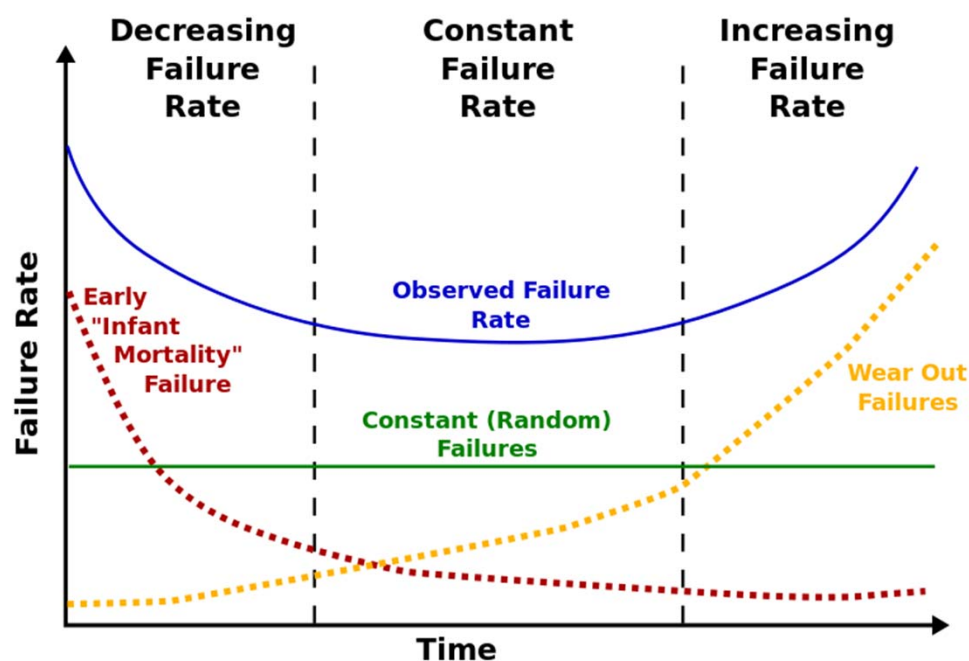
Definitions

Hazard Function

$$h(t) = \frac{1}{N(t)} \cdot \frac{\Delta n(t)}{\Delta t} = \frac{f(t)}{R(t)}$$

$\Delta n(t)$ = Number of failures in time interval $[t, t+\Delta t]$
 Δt = Length of time interval
 $N(t)$ = Population surviving at time t

$h(t)$ is the instantaneous failure rate at age t , that is, in the short time Δt from t to $t+\Delta t$, a proportion $\Delta t \cdot h(t)$ of the population *that reached age t* fails.



Definitions

Weibull Parameter Estimation

- Fitting the hazard curve allows its extrapolation in time beyond the age of the oldest assets in the population.
- The shape of the hazard curve determines which continuous distribution can be fitted to the data. Most renewal failure data sets encountered in the maintenance environment can be fitted with the Weibull distribution.

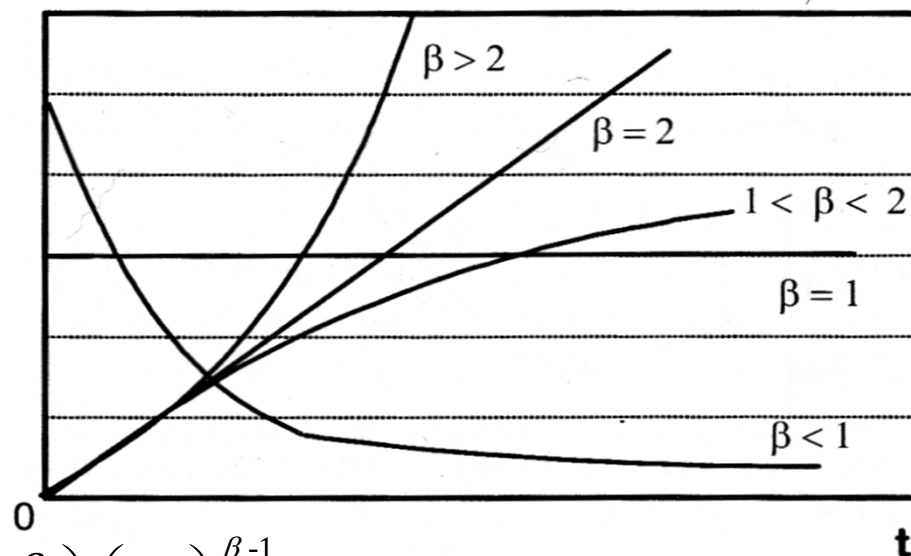
$$\text{Hazard curve: } h(t) = \left(\frac{\beta}{\alpha}\right) \cdot \left(\frac{t}{\alpha}\right)^{\beta-1}$$

$$\text{PDF: } f(t) = \left(\frac{\beta}{\alpha^\beta}\right) t^{\beta-1} \exp\left[-(t/\alpha)^\beta\right] \quad ; \quad t > 0$$

- The parameter α is called the “*characteristic life*”, since it is always the $100 \times (1 - e^{-1}) \cong 63.2th$ percentile. α has the same units as t , for example, year.
- The parameter β is called the *shape parameter* and is positive. β is dimensionless pure number and determines the shape of density function.

Definitions

Weibull Parameter Estimation



Hazard curve:
$$h(t) = \left(\frac{\beta}{\alpha}\right) \cdot \left(\frac{t}{\alpha}\right)^{\beta-1}$$

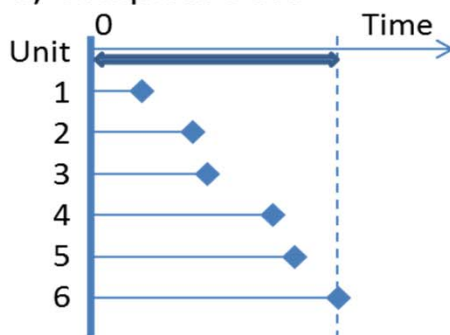
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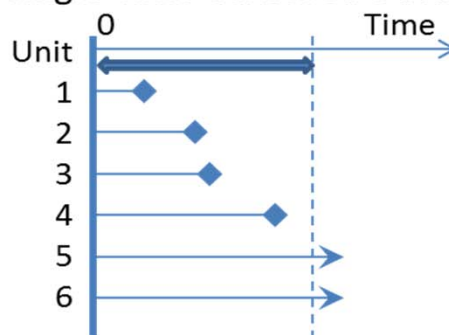
Definitions

Censoring of Statistical Data

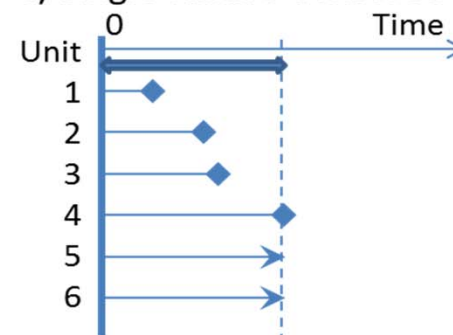
a) Complete Data



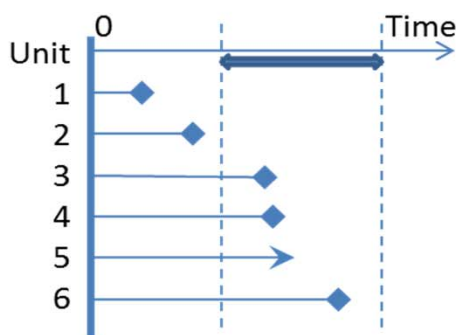
b) Single Time Censored Data



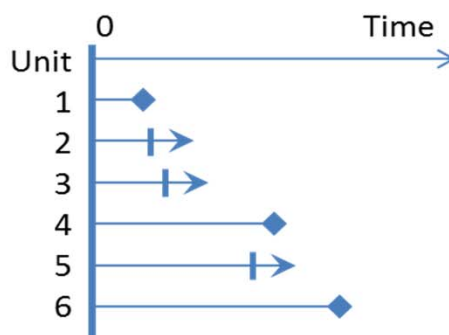
c) Single Failure Censored Data



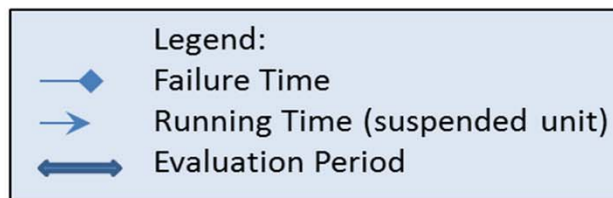
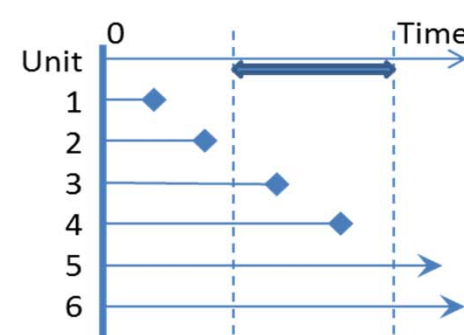
d) Left Censored Data



e) Multiply Time Censored Data



f) Left and Right Censored Data



- Canada
- Germany

DESCRIPTION OF EXISTING TRANSFORMER RELIABILITY SURVEYS

Canadian Electricity Association

- Data contribution from 11 utilities, report is published annually and is based on data of the period of the **last 5 years**.
- The survey covers transmission equipment (transmission lines, transformer banks, circuit breakers, cables) in Canada with an operating voltage of 60 kV and above.
- Collection of minor and major failures

Voltage Classification	Transformer Banks	Shunt Reactor Banks
Up to 109 kV	1,090	67
110 – 149 kV	2,327	7
150 – 199 kV	154	1
200 – 299 kV	1,299	22
300 – 399 kV	583	40
500 – 599 kV	100	68
600 – 799 kV	837	689

Canadian Electricity Association Power Transformers



Voltage Classification	Terminal Years (a)	Number of Outages	Total Time (h)	Frequency (Per a)	Mean Duration (h)
Up to 109 kV	6,079	182	125,917	0.0299	691.9
110 - 149 kV	10,528	743	243,408	0.0706	327.6
150 - 199 kV	729	52	17,231	0.0714	331.4
200 - 299 kV	6,002	407	140,790	0.0678	345.9
300 - 399 kV	2,321	172	110,588	0.0741	643.0
500 - 599 kV	507	100	26,814	0.1974	268.1
600 - 799 kV	3,205	134	161,194	0.0418	1,202.9



SUBCOMPONENT	NUMBER OF OUTAGES	FREQUENCY (PER a)	TOTAL TIME (h)	MEAN DURATION (h)	MEDIAN DURATION (h)	MEAN OP.POS. (h)
Other	67	0.0112	33,945	0.0	30.62	506.4
Bushings (Including C.T.'s)	104	0.0173	6,313	0.0	22.23	60.7
Windings	6	0.0010	10,164	0.0	1,395.07	1,694.0
On-Load Tap Changer	60	0.0100	8,426	0.0	19.94	140.4
Core	0		0	0.0		
Leads	1	0.0002	12,257	0.0	6,128.71	12,257.4
Cooling Equipment	32	0.0053	6,236	0.0	9.52	194.9
Auxiliary Equipment	46	0.0077	3,175	0.0	2.43	69.0
All Integral Subcomponents	316	0.0527	80,516	254.8	19.57	254.8

German FNN-Statistik

Objective



- Compilation of reliable statistic about number, *mode* and cause of failures of equipment in the network
- It includes about 50% of medium and high voltage grid.
- Data volume of about 4393 transformers (110, 220, 380 kV)
- Data collection by trained staff
- Catalog of code numbers for automatic data processing
- Publication of annual report

- Objective:
 - 1) Systematic collection of data on *disturbances* of primary components of the electrical grid.
 - 2) Systematic collection of data on *availability* of the electrical power supply.

- Since 2004 VDN-statistic contains no asset management data



Recording Scheme (1/2)

Quantitative network data

Quantity structure of network equipment

- Stations
- Transformers
- Switchgear
- Circuits (Overhead lines / cables)

Data on failure events

1 Organizational Information

- Identification of network operator
- Identification of network by number
- Identification of failure event

2 Network description by characteristics

- Voltage level
- Method of neutral point connection
- Network type
(share of overhead line / cables)

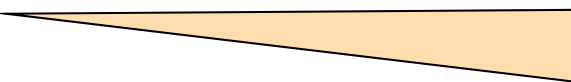
Recording Scheme (2/2)

3 Description of failure occurrence

- Time of occurrence
- Occasion of failure
- Effect of failure (e.g. protection tripping, manual disconnection)
- Type of failure (e.g. earth fault, short circuit, overload, repercussion)
- Correlations to failures in other networks in case of repercussion

4 Description of failure locations

- ***Failure location by type of equipment***
- *Mode of failure*
- Time to repair
- Occurrence of damages
- Cause of failure (optional)



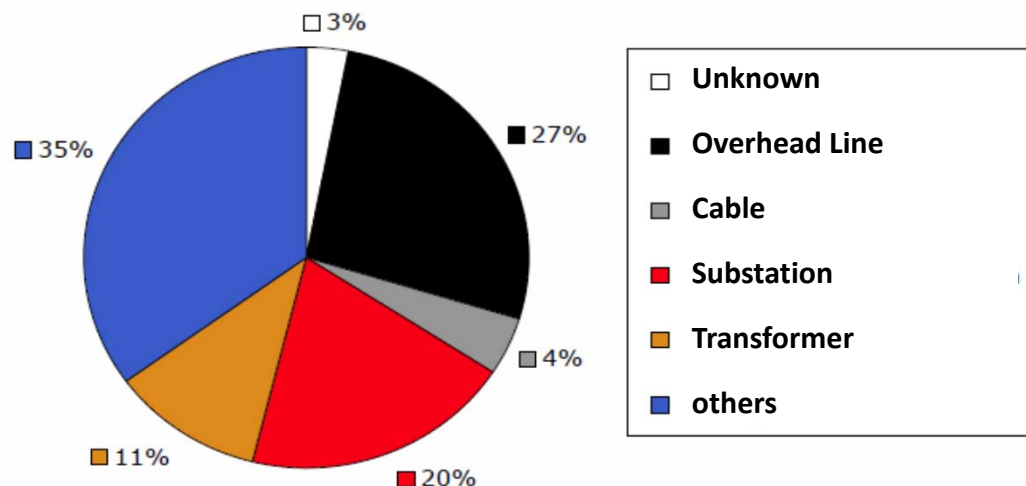
Automatic Tripping
Manual Tripping
Autoreclosure in case of
successive tripping
Common Mode Failure

5 Description of Supply Interruptions

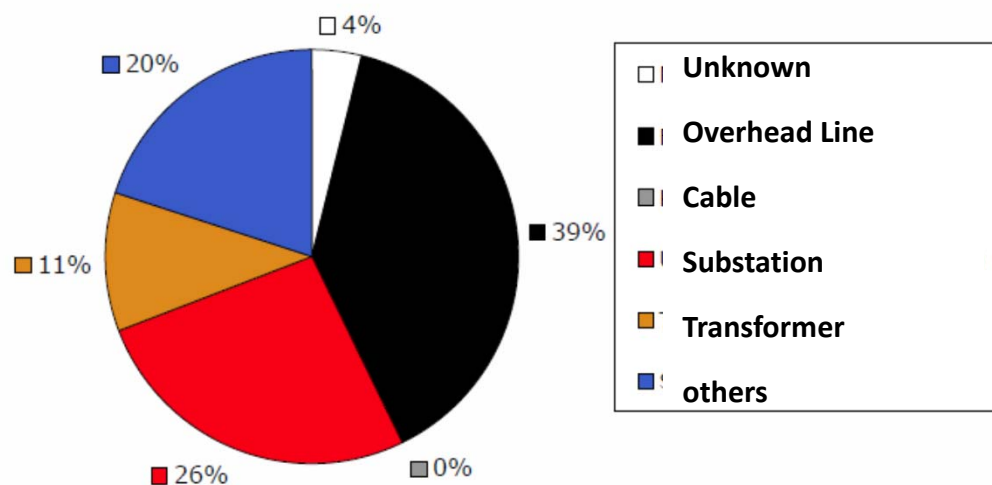
- Begin of supply interruption
- End of supply interruption
- LV: interrupted final consumers (FC) and cumulated FCmin
- MV: interrupted installed rated complex power of MV/LV-transformers (in MVA) and cumulated MV Amin
- HV/EHV: interrupted active power flow at transformers

Percentage of primary location of disturbance

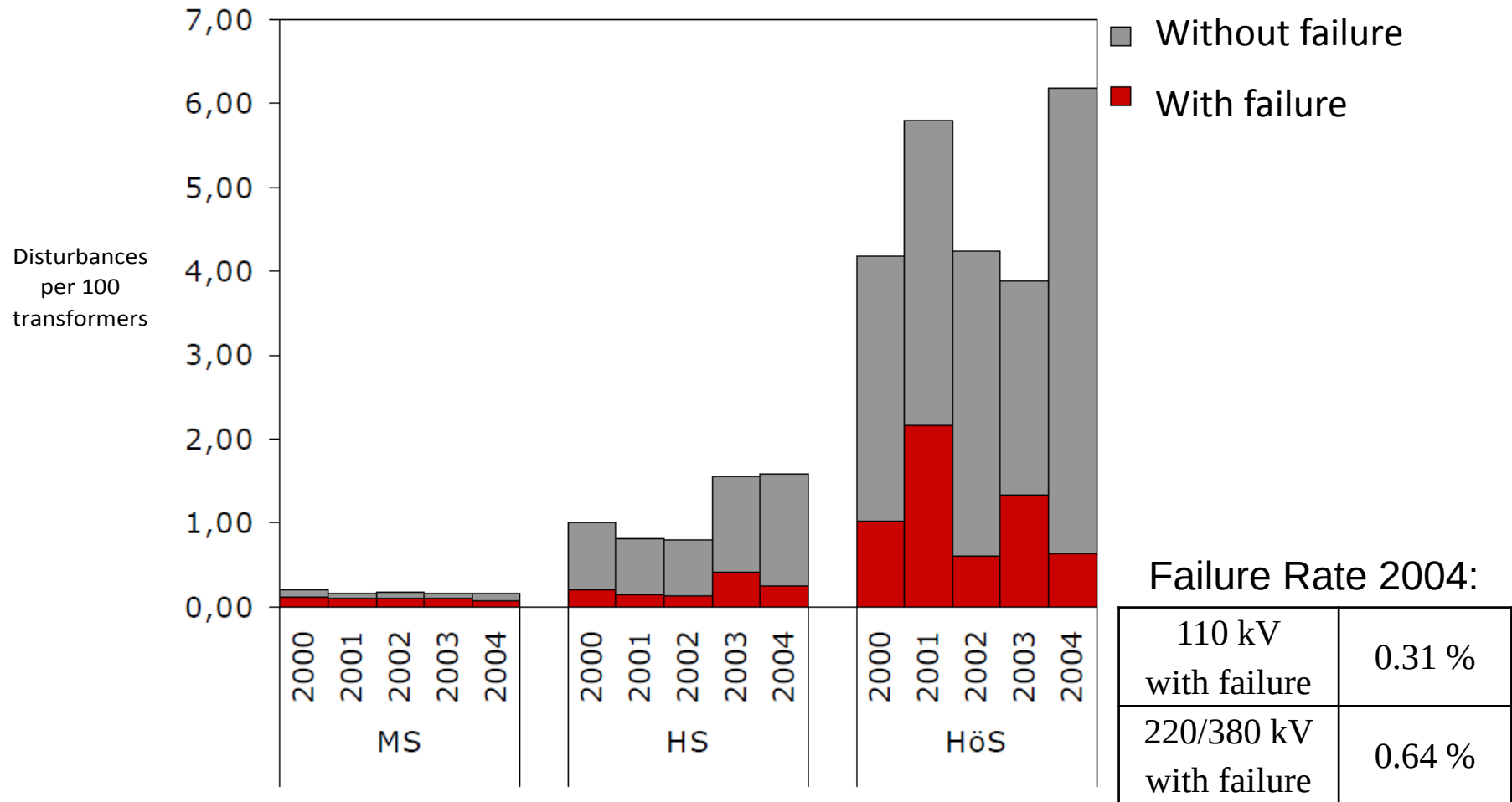
110 kV



220 kV / 380 kV



Number of disturbances per 100 transformers



Conclusion on German FNN-Statistic

- Focus on network operation
- Reliability of power supply
- For Asset Management not detailed enough, no information about
 - Type and age of equipment,
 - Location within transformer (winding, bushing, etc.)
 - Causes of failure (only deficiency, system event, environmental)
 - Loading, operational condition

STANDARDIZED FAILURE DATA COLLECTION

Motivation for CIGRE Questionnaire

- The main objective of the countrywide surveys is the systematic collection of data on the availability and disturbances of the electrical power supply, with emphasis on the frequency, duration and extent of the interruptions.
- Detailed statistics about the failure location in the respective equipment, failure cause or mode and repair activities are normally not included.
- The benefit of these statistics with respect to asset management is therefore limited.
- Different definitions and information content constrain forming a coherent database from individual sources.
- Therefore data collection is extended to utility data by means of a *questionnaire*.

See Cigre A2-Website!

www.uni-stuttgart.de/ieh/wga237.html

Definition of Major Failure

- Any situation which requires the equipment to be removed from service for a period longer than 7 days for investigation, remedial work or replacement is a major failure.
- Where repairs are required, these involve major remedial work, often requiring the transformer to be removed from its plinth and returned to the factory.
- A major failure would require at least the opening of the tank, including the tap changer tank or an exchange of bushings.
- Also a reliable indication that the condition of the transformer prevents a safe operation should be counted as a major failure if remedial work (longer than 7 days) is needed for restoring original service capability (e.g. detection of strong PDs).

Questionnaire – Population Data

3.1 Transformer application:

- Substation – Distribution,
- Substation – Transmission,
- Power Station – Generator Step-Up,
- Power Station – Unit Transformer
- Shunt Reactor
- Other

3.2 Type:

- Two Winding,
- Two Winding with Tertiary,
- Autotransformer,
- Autotransformer with Tertiary

3.3 Number of phases in tank

- 1-Phase,
- 3-Phases

3.4 Voltage Ratios dependent on application and highest system voltage

3.5 Rated power dependent on application and highest system voltage

3.6 Typical loading dependent on application and highest system voltage

3.7 Typical manufacturing period dependent on application and highest system voltage.

But no distinct age distribution!

Questionnaire – Failure Data

- *Identification of the unit:* application, type, construction type, year of manufacture.
- *Features of the unit:* rated power, rated voltage, number of phases, cooling system, type of oil, tap changer, tap changer arrangement, oil preservation system, over voltage protection.
- *Detail of occurrence:* year of failure, service years to failure, loading immediately prior to failure.
- *Consequences of failure:* external effects, failure location, service years of failed bushings (if location is bushings), failure mode, failure cause, action taken, and detection mode.

Questionnaire – Failure Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	This survey should cover major failures only!															
2	1 - IDENTIFICATION OF THE UNIT					2 - FEATURES OF THE UNIT									3 - DE	
3	1.1	1.2	1.3	1.4	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.1	3.2	
4	No.	Application	Type	Construction Type	Year of Manufacture	Rated Power [MVA]	Nominal Voltage [kV]	Number of Phases	Cooling System	Type of Oil	Tap Changer	Tap Changer Arrangement	Oil Preservation System	Over Voltage Protection	Year of Failure	
5	1	Substation - Distribution	Two Winding with Tertiary	Core Type	1974	100	150	Three Phase	ONAF	Mineral Oil	On-Load Tap-Changer (oil breaker)	Combined tap selector and diverter switch in separate compartment from main tank	Free breathing	External Surge arresters	1980	
6	2	Substation - Distribution	Two Winding with Tertiary									Combined tap selector and		External Surge		
7	3	Substation - Distribution	Two Winding	3 - DETAIL OF OCCURENCE			4 - CONSEQUENCES OF FAILURE						5 - REMARKS, Internal Identification, Repair History			
8	4	Substation - Distribution	Two Winding	3.1	3.2	3.3	4.1	4.2	4.3	4.4	4.5	4.6				
9	5	Substation - Distribution	Two Winding	Year of Failure	Service Years to Failure	Loading Immediately Prior to Failure	External Effects	Failure Location	Service Years of Failed Bushings in 4.2	Failure Mode	Failure Cause	Action				
10	6	Substation - Distribution	Two Winding	1980	6	Not known	None	Tap Changer		Physical chemistry (including oil, corrosion, contamination etc)	Design	Repair in workshop				
11	7	Substation - Distribution	Two Winding	1980	6	Not known	None	Tap Changer		Physical chemistry (including oil, corrosion, contamination etc)	Design	Repair in workshop				
12	8	Substation - Distribution	Two Winding	1987	19	Not known	None	Tap Changer		Physical chemistry (including oil, corrosion, contamination etc)	Design	Repair in workshop				
13	9	Substation - Distribution	Two Winding	1996	24	Not known	Leakages	Tank		Mechanical	Manufacturing	Onsite Repair < 1 week				

www.uni-stuttgart.de/ieh/wga237.html

See Cigre A2-Website!

Questionnaire – Failure Data

- Answers are standardized by means of a pull-down menus
- Confidentiality:
 - Origin of data is only known to collector and convenor
 - Within WG data are handled anonymously
 - Outside WG only consolidated data will be presented

- Investigated Population
- Failure Rate

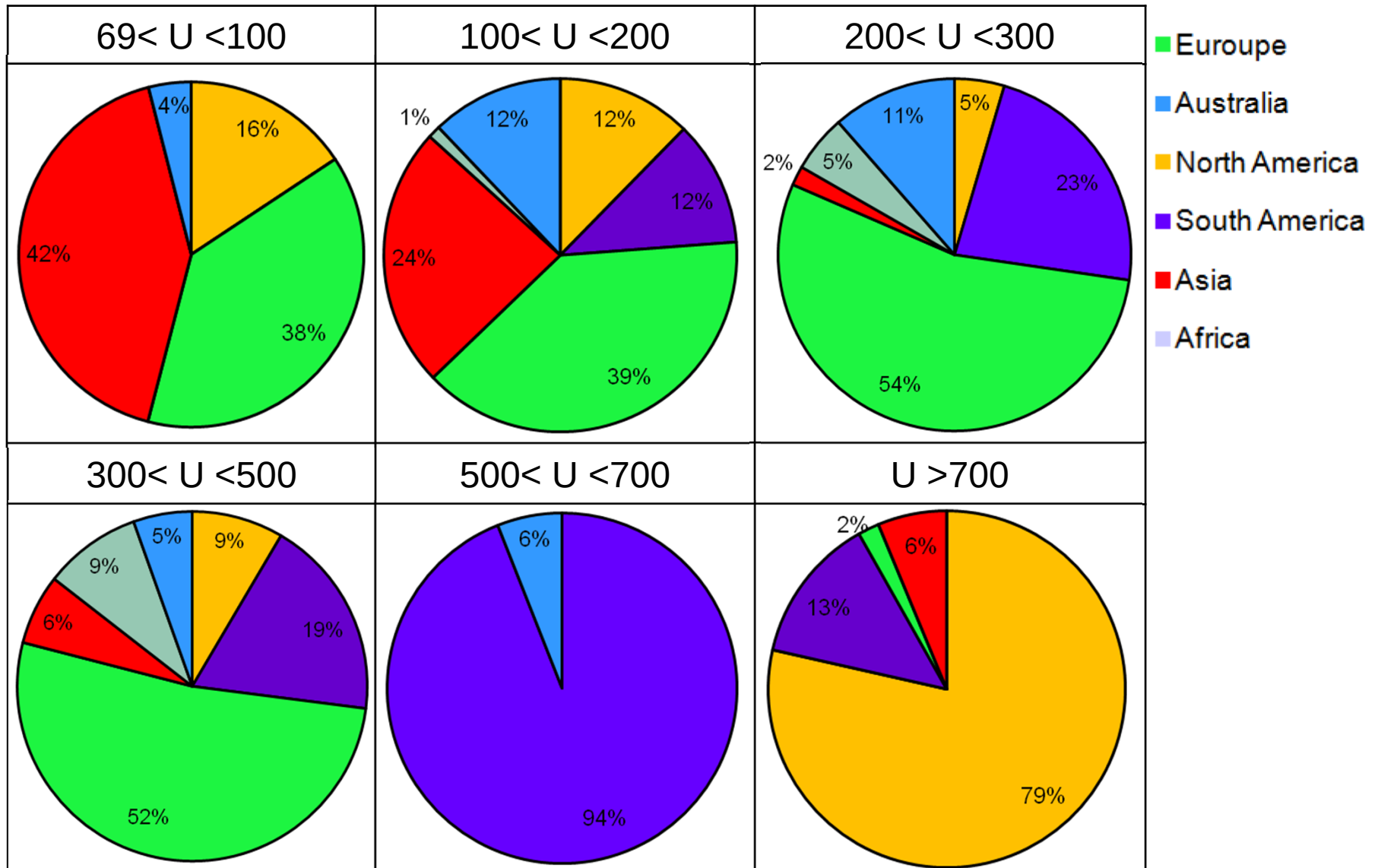
RESULTS OF RELIABILITY SURVEY

Investigated Population

- 56 utilities from 21 countries submitted failure data
- The reference periods range from 3 to 11 years.
- Population of 23.884 transformers and 167.459 transformer-years
- The year of manufacture of the units span from the 1950's up to 2009.
- Collection of 964 major failures which occurred in the period 1996 to 2010

Average reference period:
$$\bar{T} = \frac{Tf - Years}{NumberTf} = 7.95 years$$

Investigated Population dependent on Voltage Class and Origin



Investigated Population dependent on Application



$$\bar{T} = \frac{Tf - \text{Years}}{\text{Number}Tf} = 6.8 \text{ years}$$

Substation Transformers

POPULATION INFORMATION	HIGHEST SYSTEM VOLTAGE [kV]						
	69 ≤ kV < 100	100 ≤ kV < 200	200 ≤ kV < 300	300 ≤ kV < 500	500 ≤ kV < 700	kV ≥ 700	All
Number of Utilities	10	38	31	27	3	4	51
Number of Transformers	2.962	10.932	4.272	3.233	434	348	22.181
Transformer-Years	15.267	64.718	37.017	25.305	4.774	2.991	150.072

$$\bar{T} = \frac{Tf - \text{Years}}{\text{Number}Tf} = 10.2 \text{ years}$$

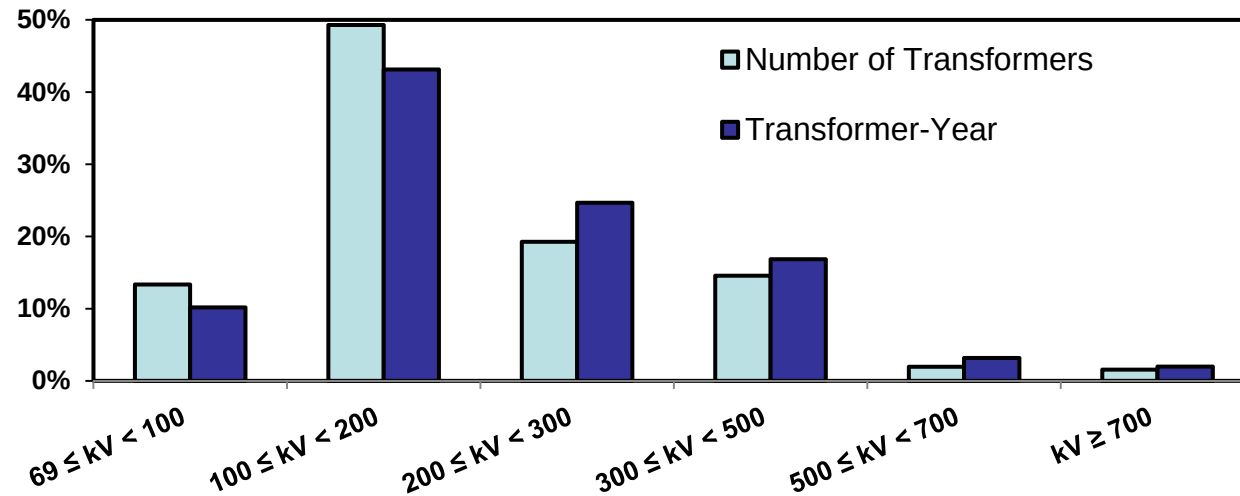
Generator Step-Up Transformers

POPULATION INFORMATION	HIGHEST SYSTEM VOLTAGE [kV]						
	69 ≤ kV < 100	100 ≤ kV < 200	200 ≤ kV < 300	300 ≤ kV < 500	500 ≤ kV < 700	kV ≥ 700	All
Number of Utilities	3	17	20	13	1	1	26
Number of Transformers	14	320	455	673	167	74	1,703
Transformer-Years	153	3,278	4,639	6,740	1,837	740	17,387

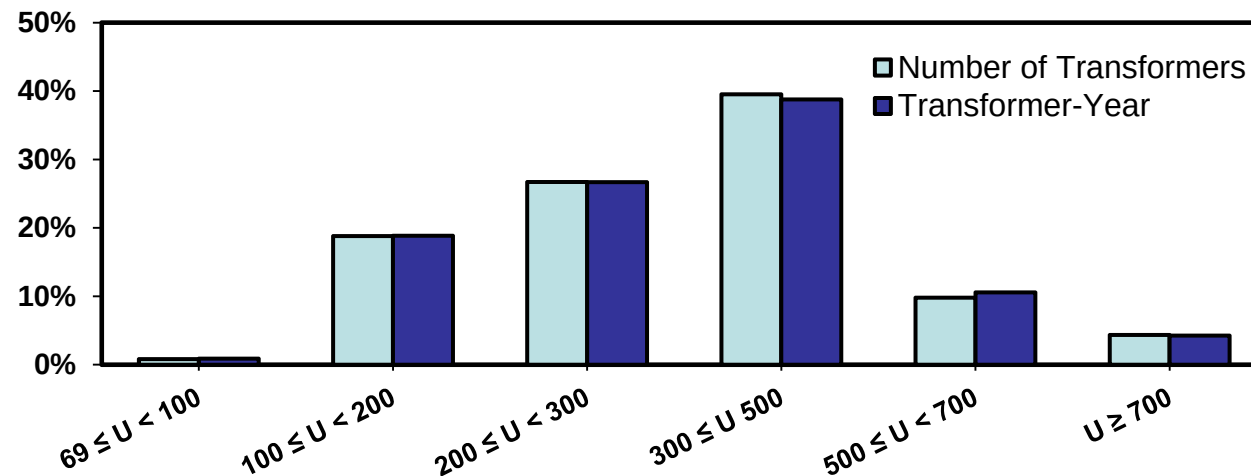
Investigated Population dependent on Application



Substation Transformers



Generator Step-Up Transformers



Failure Rate Substation Transformers



FAILURES & POPULATION INFORMATION	HIGHEST SYSTEM VOLTAGE [kV]						
	$69 \leq \text{kV} < 100$	$100 \leq \text{kV} < 200$	$200 \leq \text{kV} < 300$	$300 \leq \text{kV} < 500$	$500 \leq \text{kV} < 700$	$\text{kV} \geq 700$	All
Failures	144	280	186	152	27	10	799
Transformer-Years	15,267	64,718	37,017	25,305	4,774	2,991	150,072
FAILURE RATE	0.94%	0.43%	0.50%	0.60%	0.57%	0.33%	0.53%

Failure Rate GSU Transformers

FAILURES & POPULATION INFORMATION	HIGHEST SYSTEM VOLTAGE [kV]						
	$69 \leq \text{kV} < 100$	$100 \leq \text{kV} < 200$	$200 \leq \text{kV} < 300$	$300 \leq \text{kV} < 500$	$500 \leq \text{kV} < 700$	$\text{kV} \geq 700$	All
Failures	0	20	43	89	9	4	165
Transformer-Years	153	3,278	4,639	6,740	1,837	740	17,387
FAILURE RATE	0.00%	0.61%	0.93%	1.32%	0.49%	0.54%	0.95%

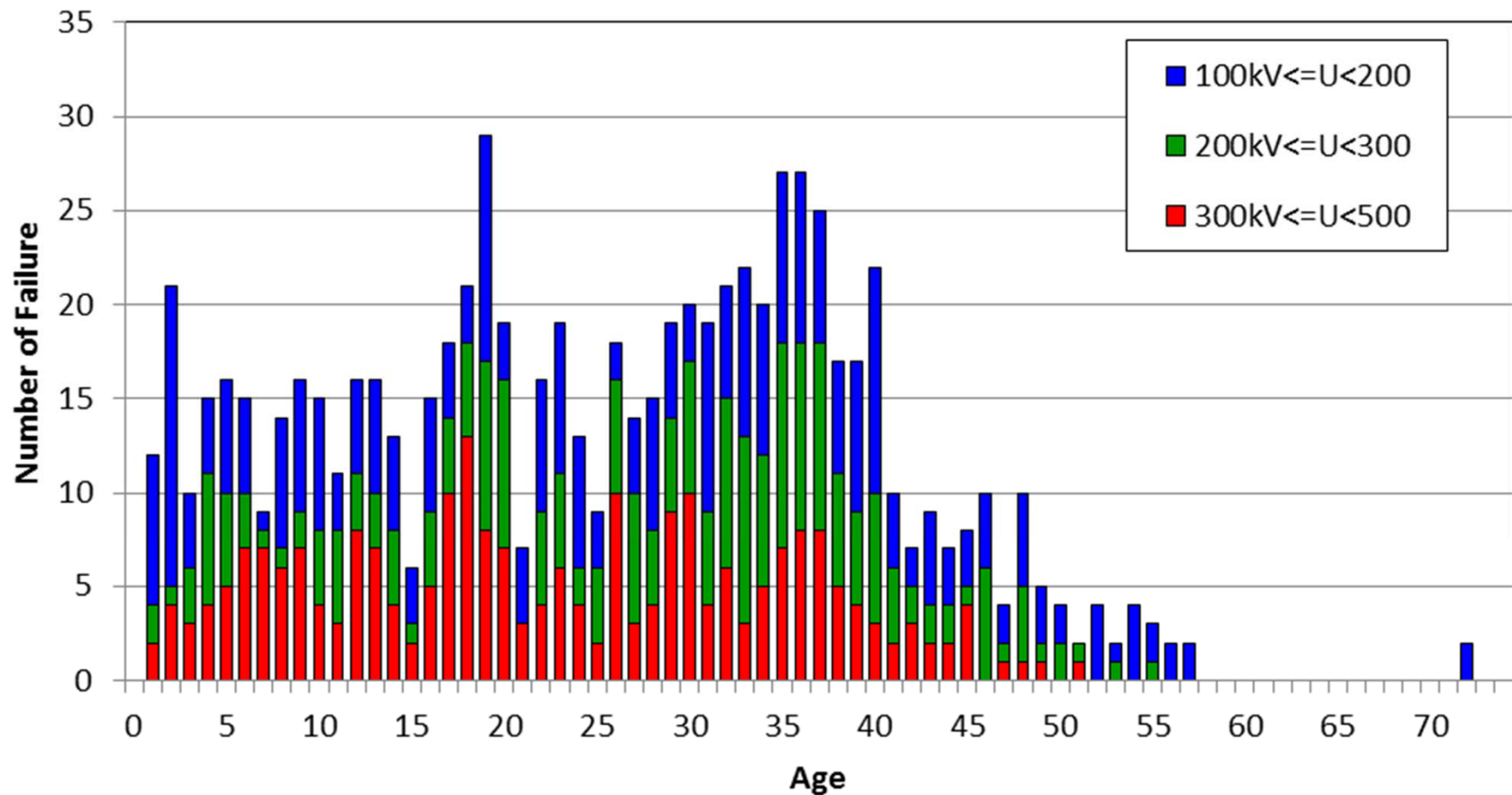
Failure Rates from other Surveys

SURVEY	APPLICATION/ CLASSIFICATION	FAILURE PERIOD	MANUFACTURING PERIOD	FAILURE RATE (%)	SOURCE
Cigré International Survey	All voltage levels (60kV-700kV)	1968 - 1978	Pre 1978	2	[Bossi, 1983]
United Kingdom	All voltage levels	Pre 1987	Pre 1987	< 2	[Allan, 1987]
United Kingdom	Generator Step-Up, Major failures	1974 - 1995	Pre 1995*	1.2	[Hall, 2006]
ZTZ Service Database, Ukraine	Generator Step-Up & Transmission Rated power $\geq 100\text{MVA}$,	2000 - 2005	Pre 2005	1 - 2	[Sokolov, 2005-2]
US-NGRID, United States	Distribution 115kV, 69kV & <69kV	-	-	0.35 - 0.8	[Prout , 2003]
Hydro Quebec, Canada	All voltage categories, major failures	-	-	< 0.5	[Foata, 2006-1]
American Electric Power	345kV & 765kV	Pre 1986	Pre 1986 *	1.3 - 2.9	[Fleeman, 2002]
American Electric Power	345kV & 765kV	Post 1986	Post 1986 *	0.35 - 1.35	[Fleeman, 2002]
Australia & New Zealand	Costly failures	Pre 1996	Pre 1996	0.4	[Austin, 2001]
* Indicates studies where the manufacturing period was given.					

- Hazard Curve
- Weibull Distribution to Predict Future Failure Behavior

RESULTS OF RELIABILITY SURVEY

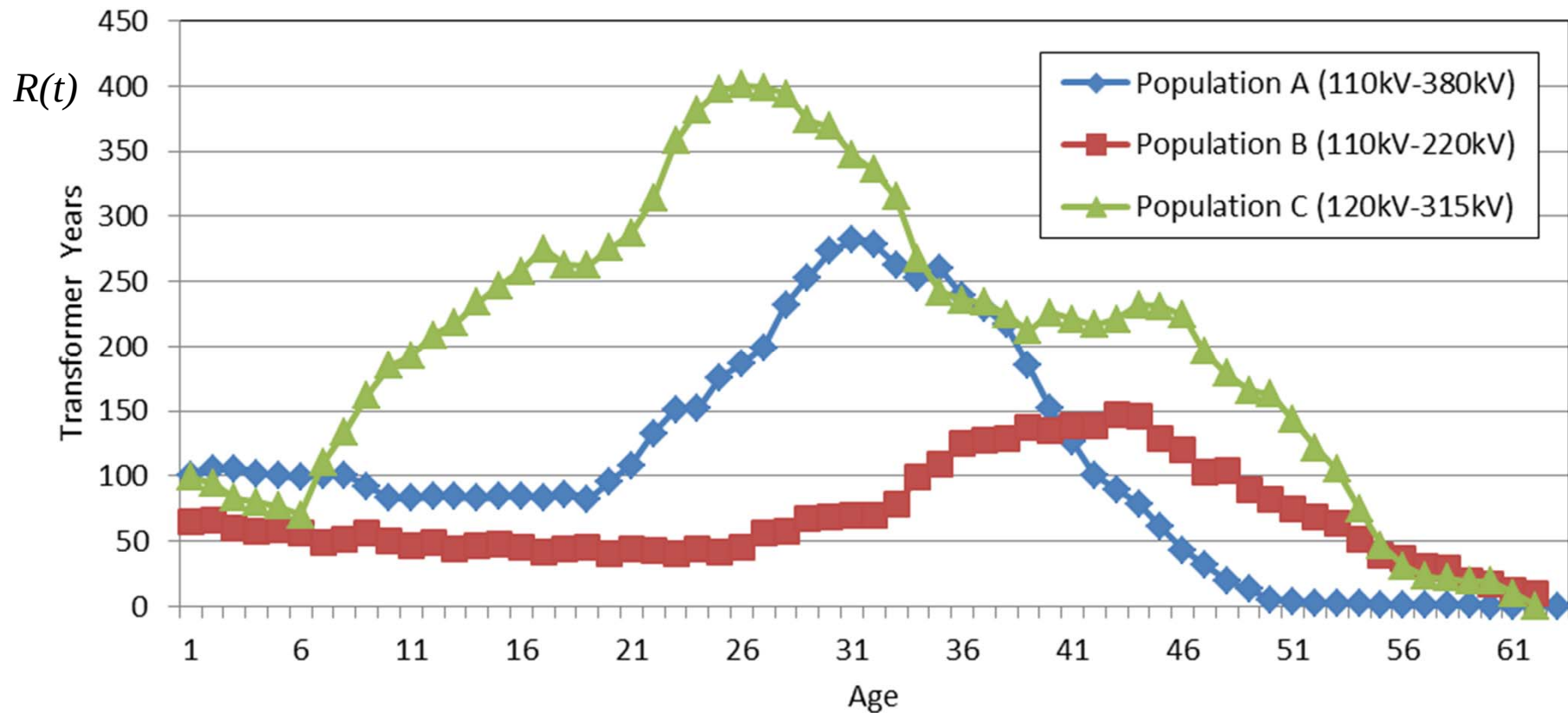
Number of Failures Dependent on Age Worldwide (100 kV – 500 kV)



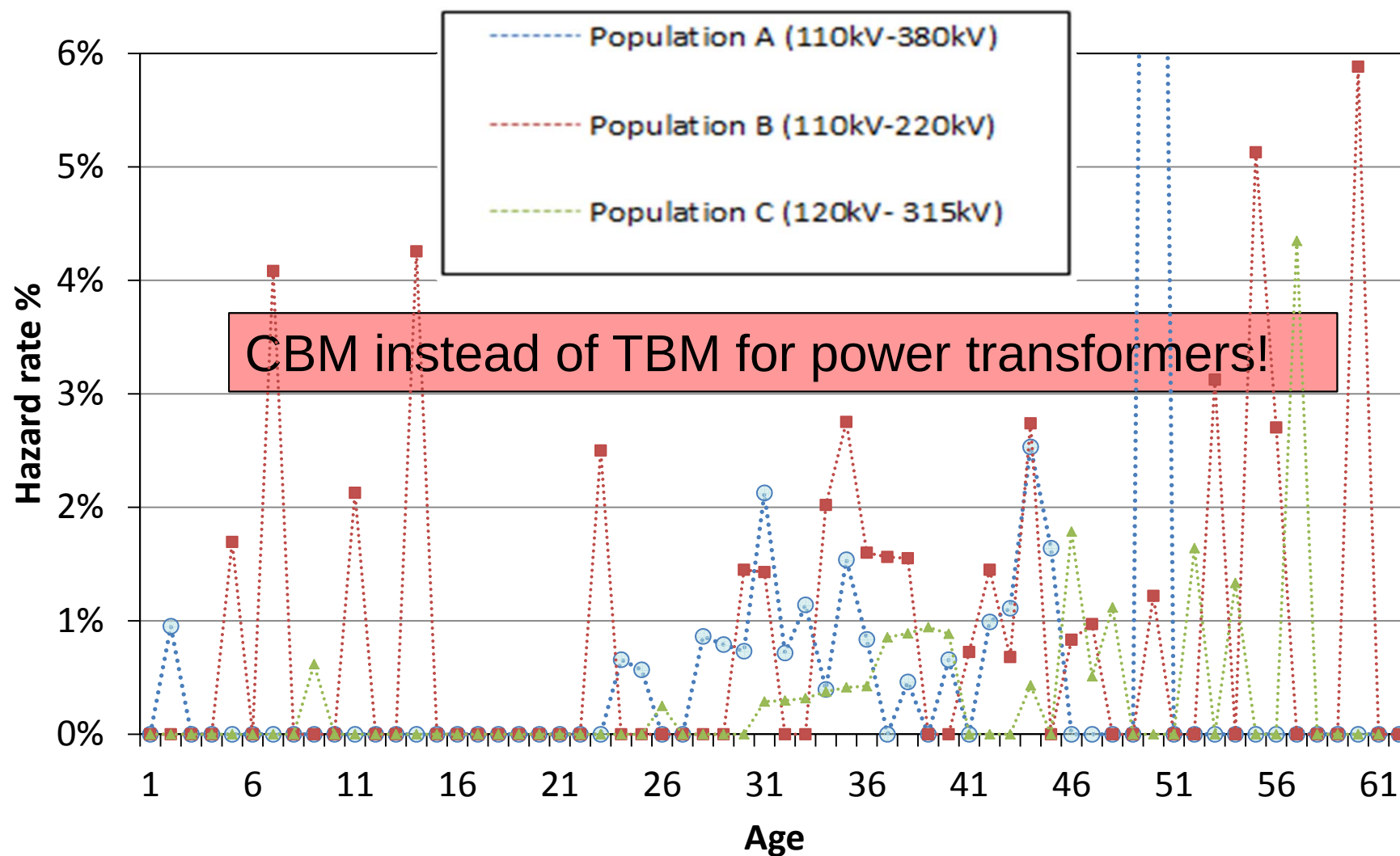
Number of Transformers-Years surviving Age T (considering 11 year failure interval)



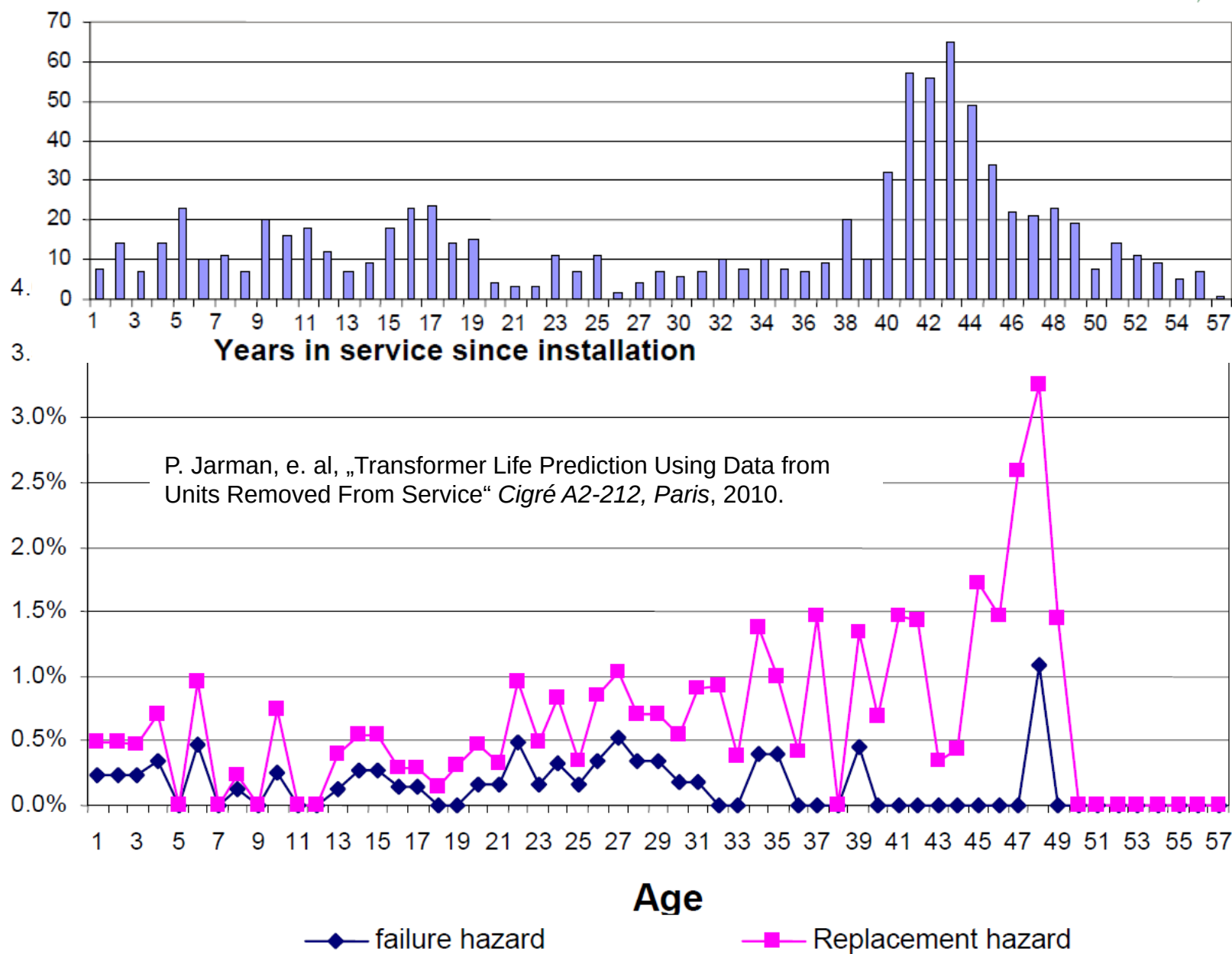
$$h(t) = \frac{1}{N(t)} \cdot \frac{\Delta n(t)}{\Delta t} = \frac{f(t)}{R(t)}$$



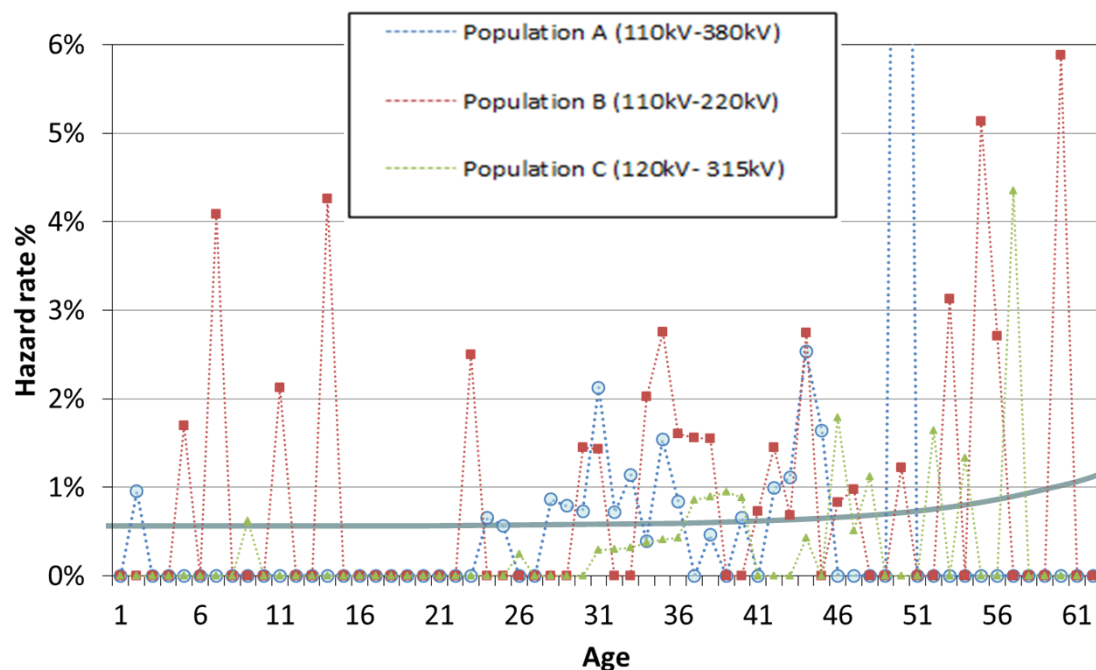
Failure Hazard Rate



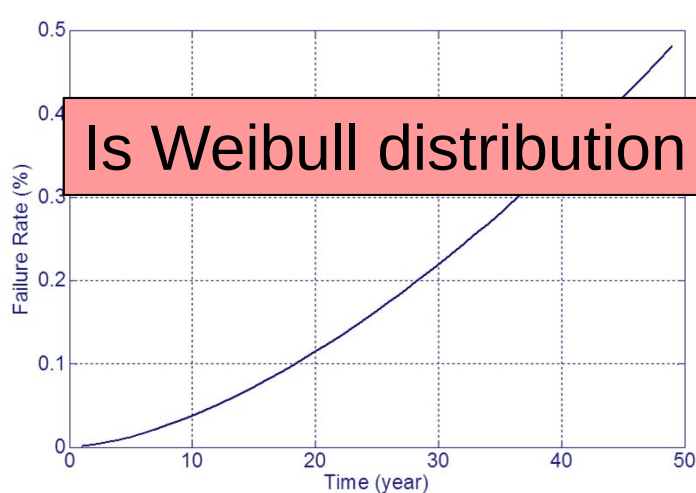
Hazard Curve – National Grid (UK)



Failure Rate Fitting using Weibull Distribution



$$h(t) = \left(\frac{\beta}{\alpha} \right) \cdot \left(\frac{t}{\alpha} \right)^{\beta-1}$$

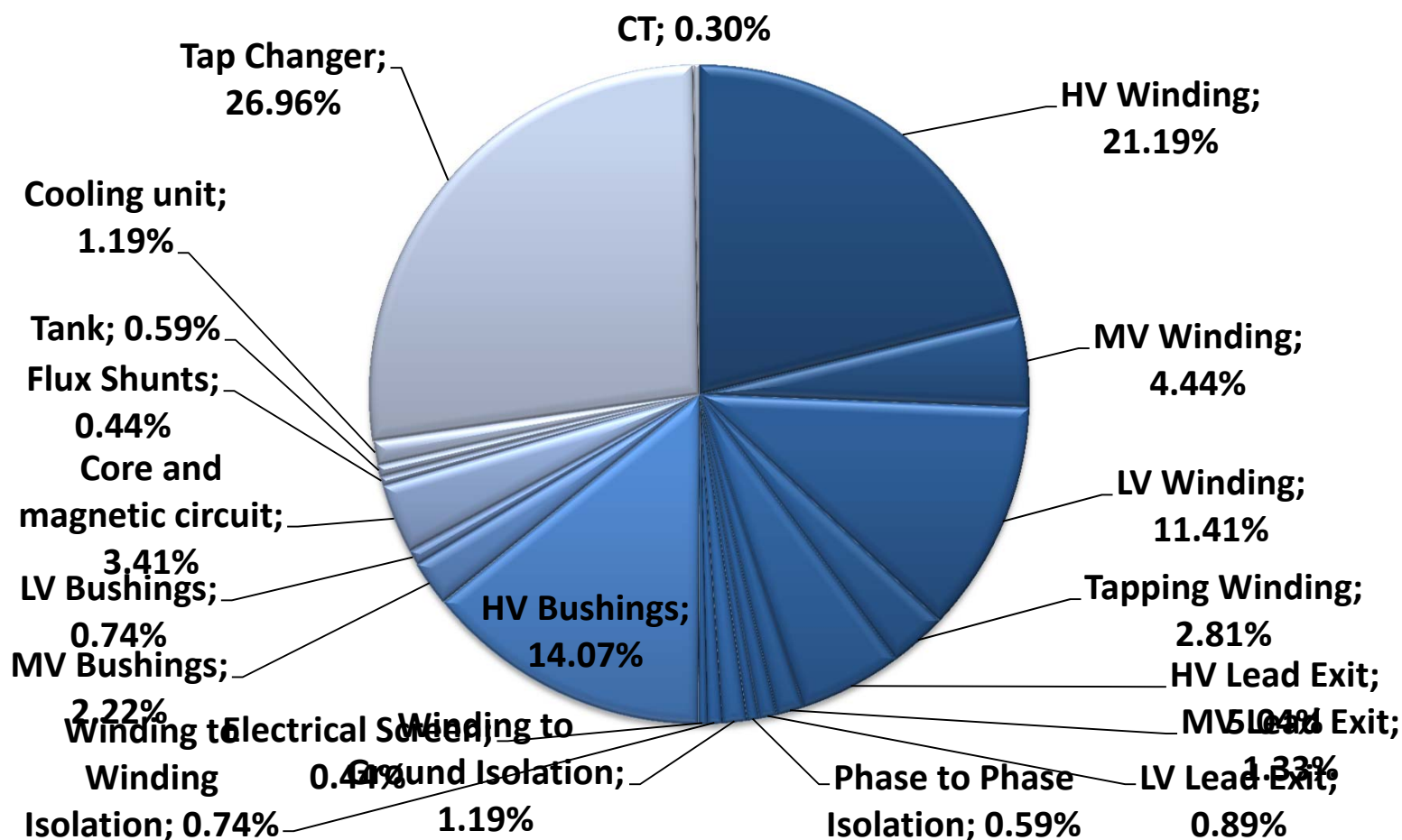


Is Weibull distribution applicable to power transformers?

- Failure Location
- Failure Mode
- Failure Cause
- External Effects

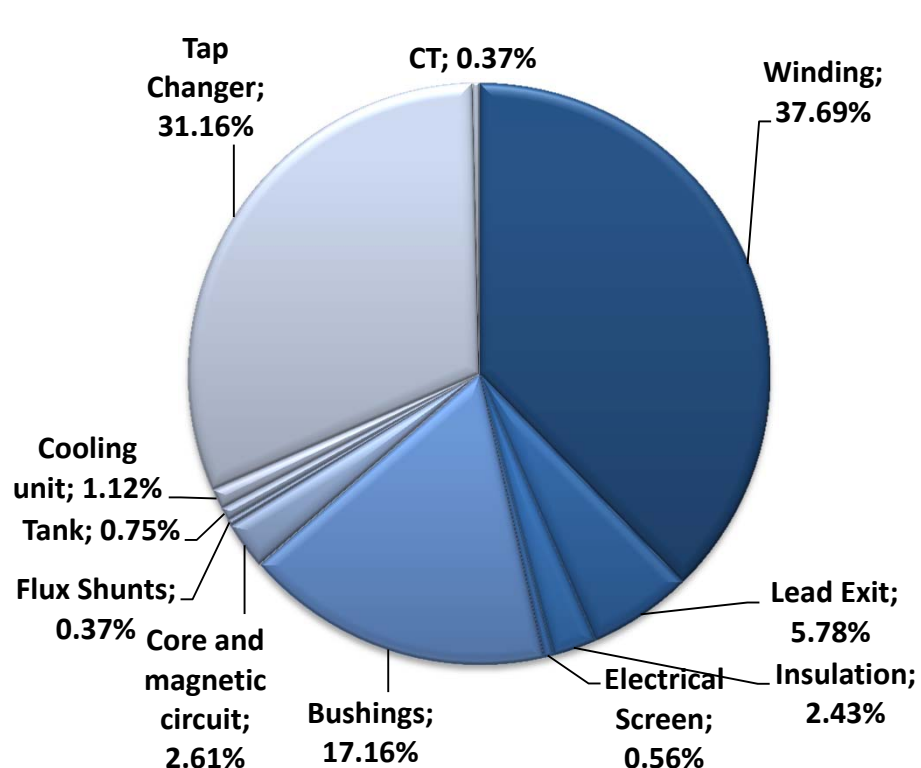
RESULTS OF RELIABILITY SURVEY

Failure Location

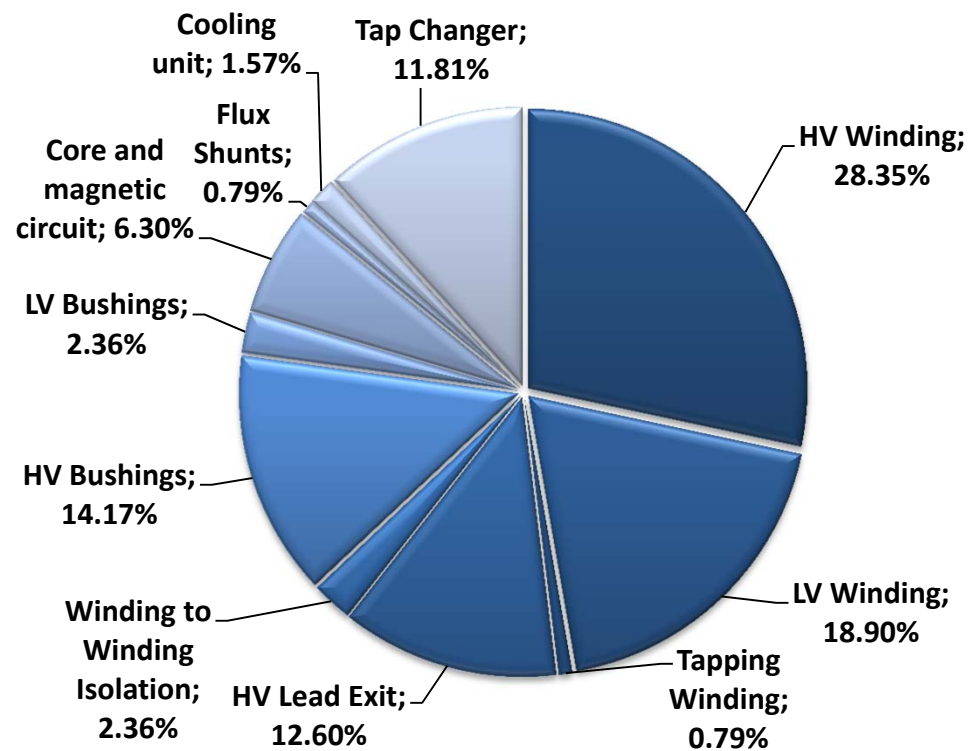


based on 964 failures

Failure Location dependent on Application (U>100kV)

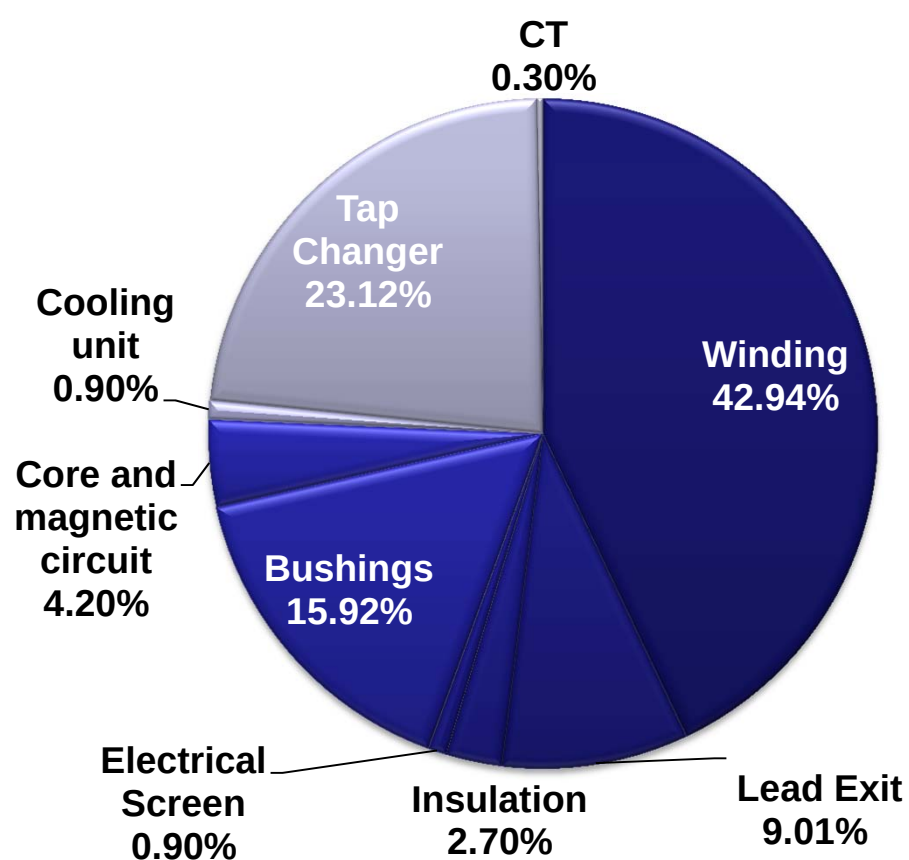


Substation Transformers
(536 major failures)

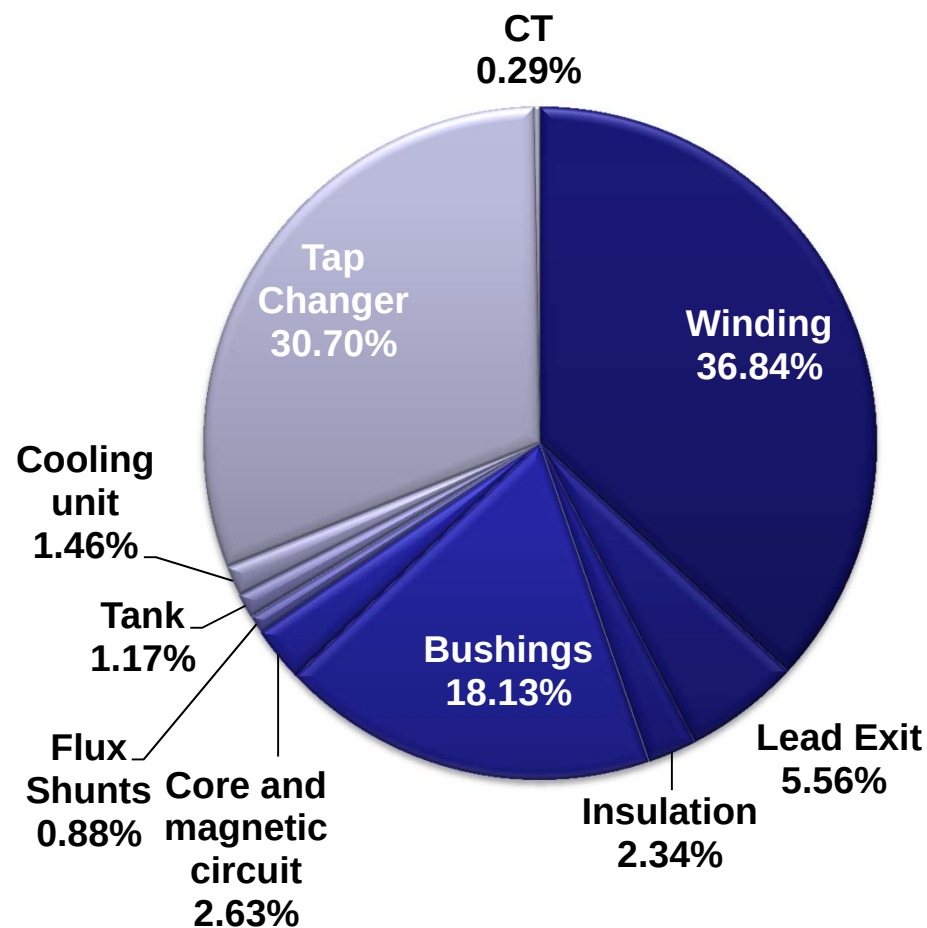


Generator Step-Up Units
(127 major failures)

Failure Location dependent on Age ($U > 100\text{kV}$)

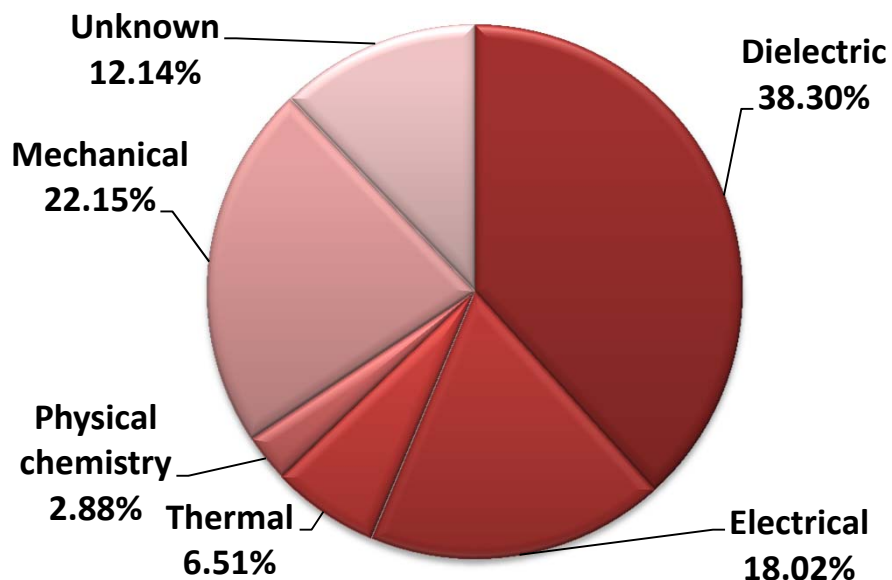


Manufacturing Period <1980
(333 major failures)

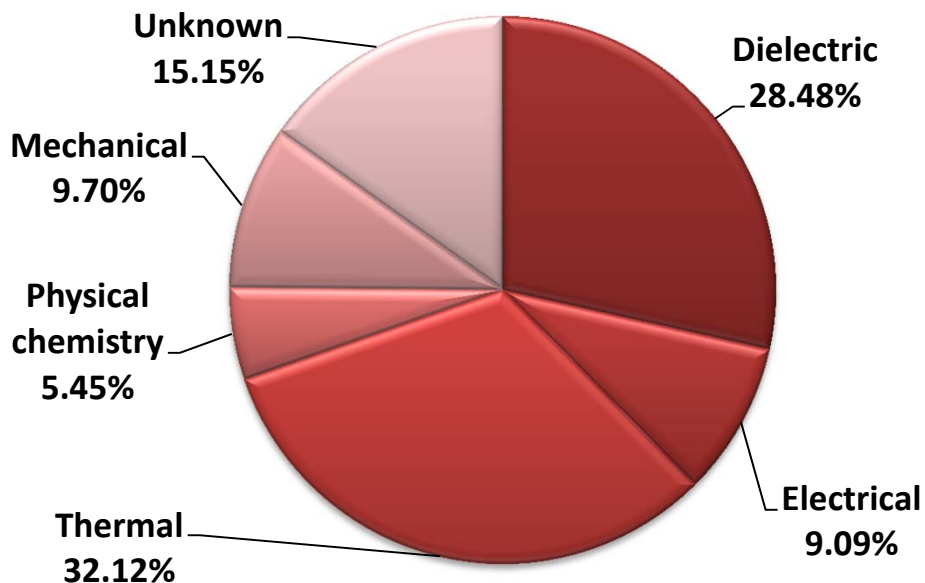


Manufacturing Period ,1980
(342 major failures)

Failure Mode

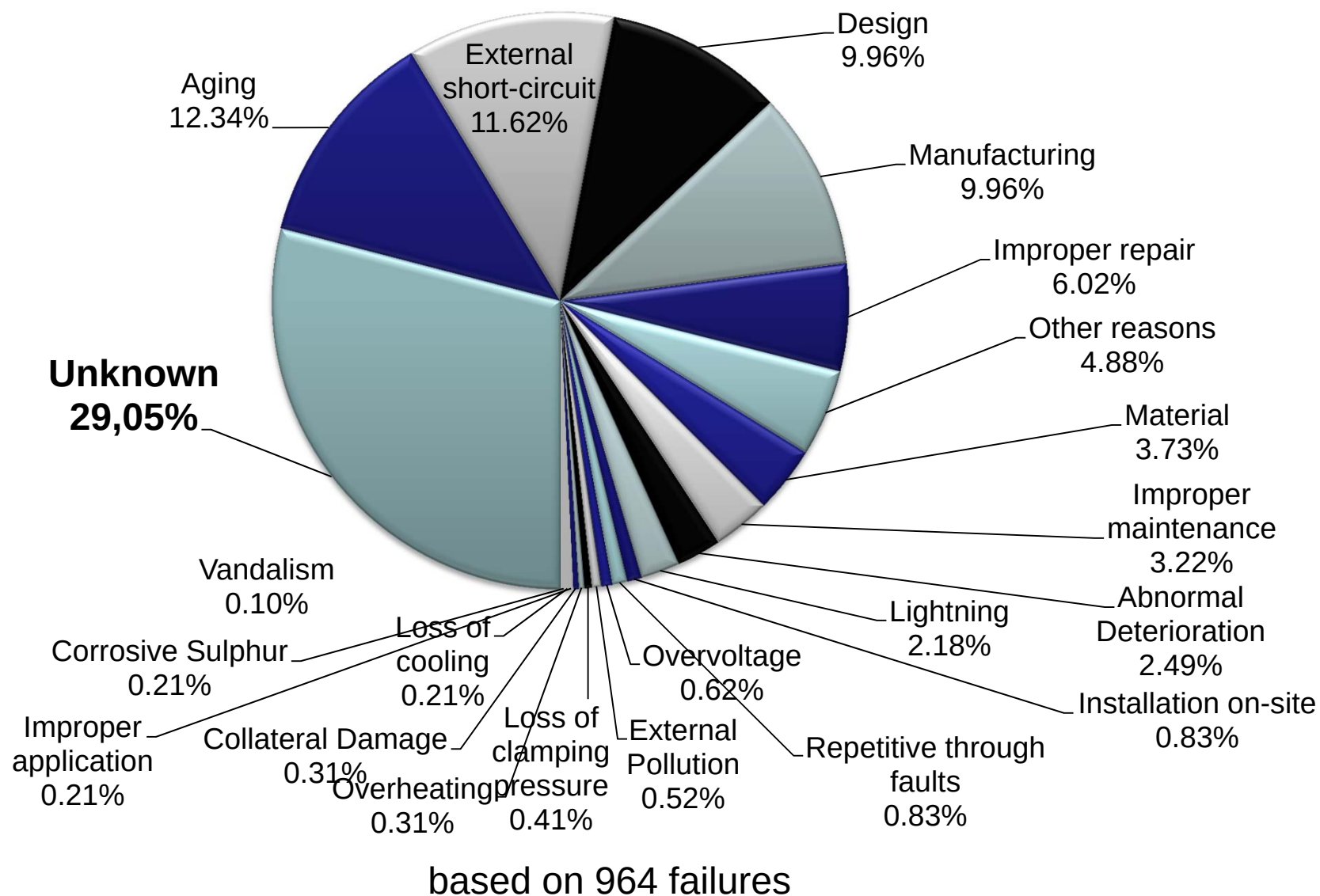


Substation Transformers
(799 major failures)

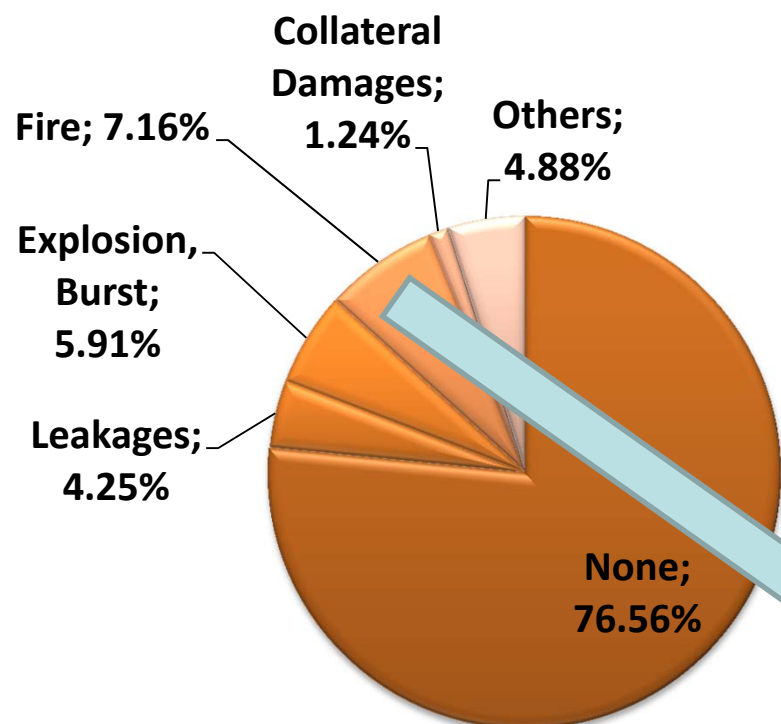


Generator Step-Up Units
(165 major failures)

Failure Cause

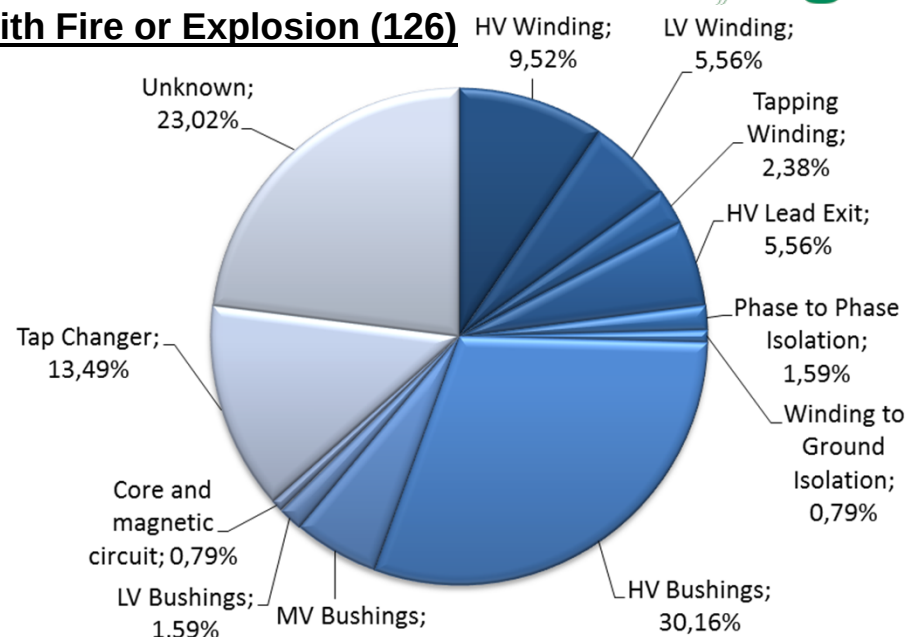


External Effects

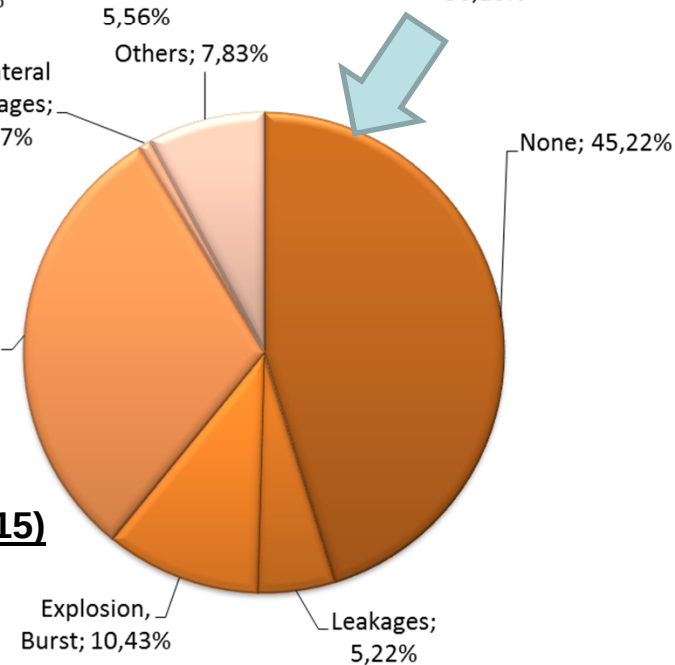


based on 964 failures

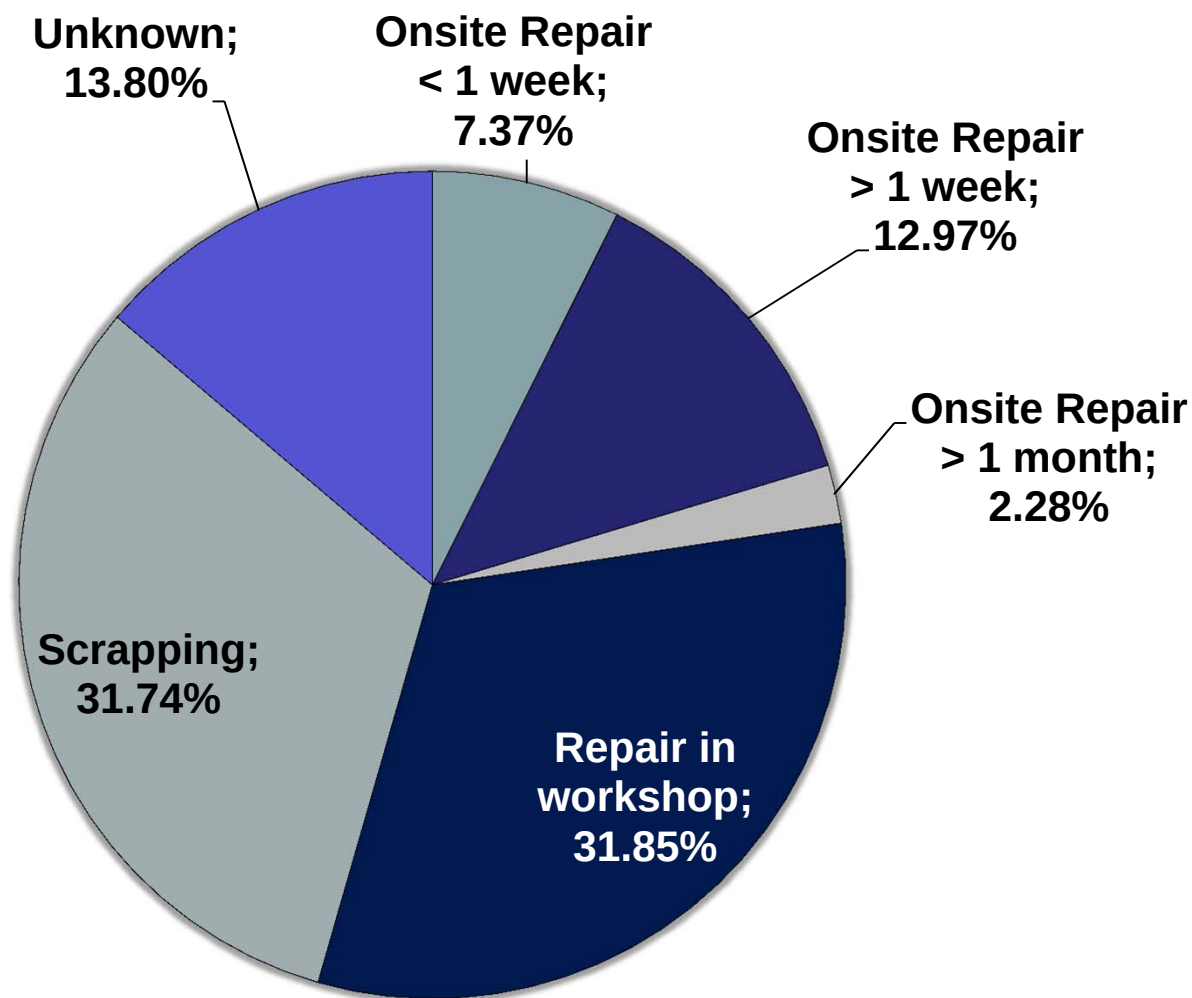
Failures with Fire or Explosion (126)



Bushing Failures (115)

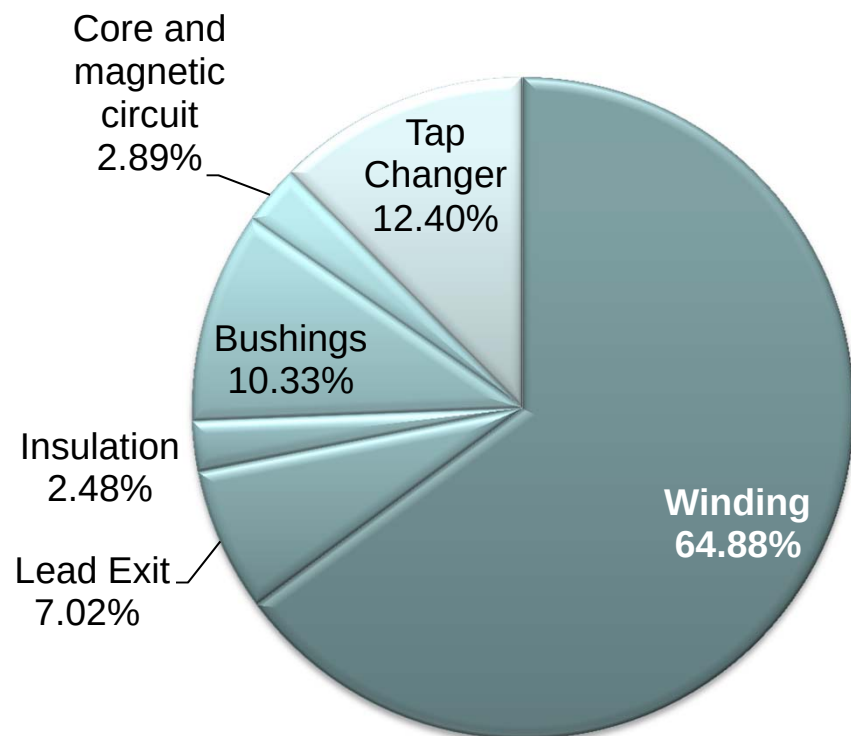


Action taken after Failure

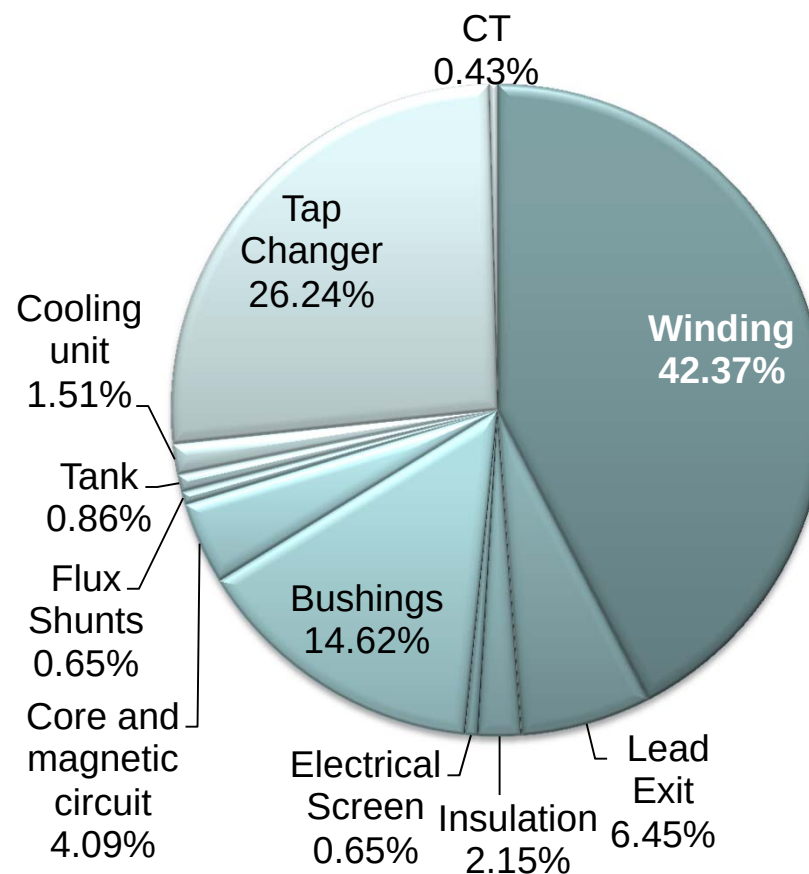


Action Dependent on Failure Location

Scrapped Transformers (242)



Repaired Transformers (465)



Conclusion and Recommendation

- Review of existing surveys for failure data acquisition: Public available statistics often focused on system reliability and not asset management
- A questionnaire was developed by which utility failure statistics can be collected in a standardized way.
- Presented results are based on a worldwide population of 23.884 transformers and 167.459 transformer-years with 964 major failures.
- Failure rate of 0.53% for substation transformers and 0.95% for GSU.
- The hazard curve function for substation transformers does not show a distinct ageing behaviour -> Top-Down analysis is not useful for predictions of individual component. CBM is necessary!
- About 50% of major failures occur in the windings.
- Bushing failures often result in fire or explosions.
- It is recommended to use questionnaire on a regular basis.